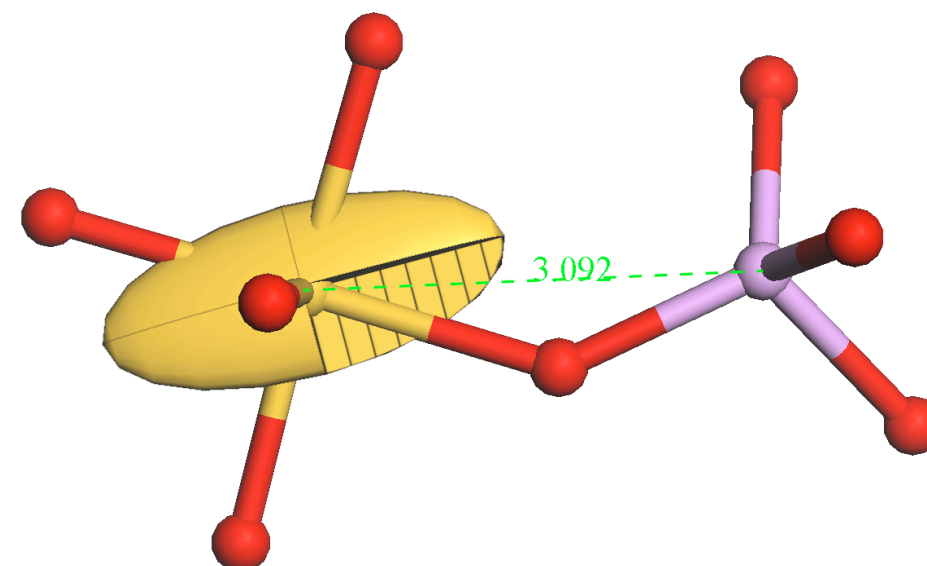
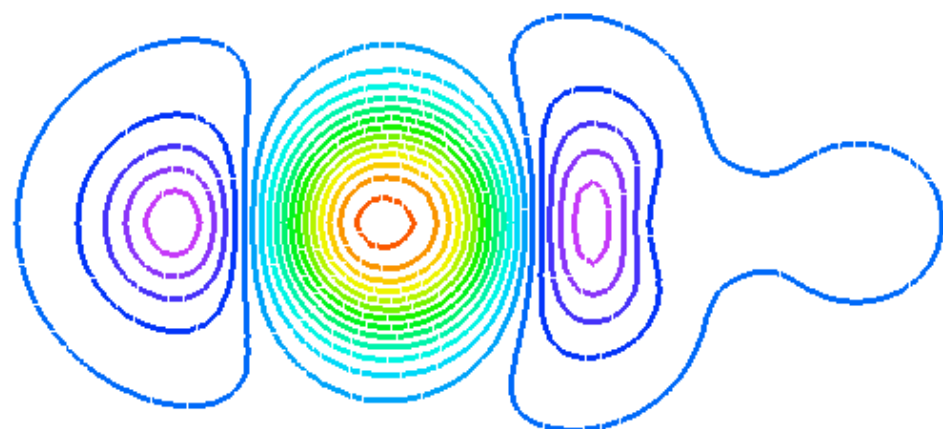


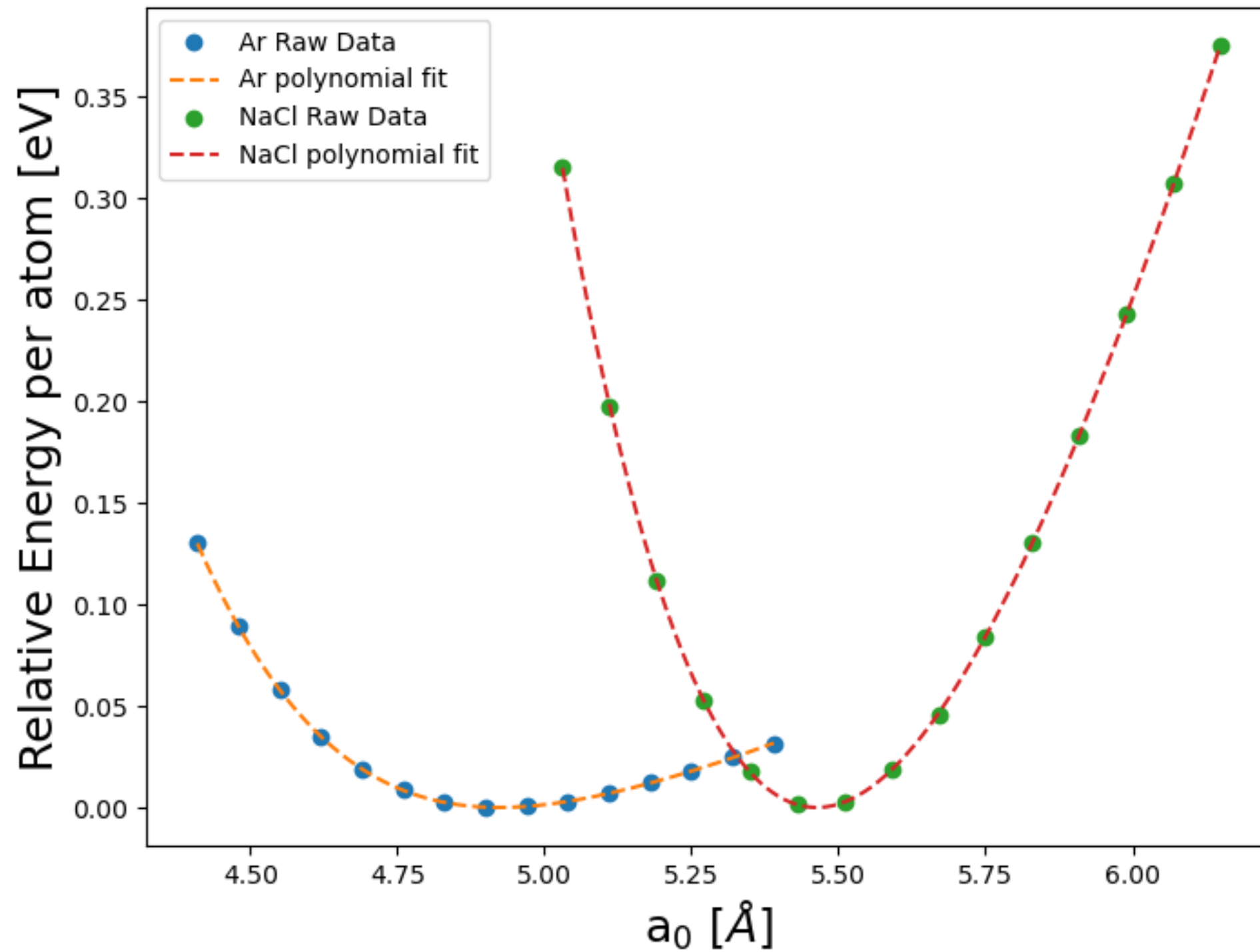
Ground-state Structure and Dynamics

Jonathan Yates

Materials Modelling Laboratory, Oxford Materials



Ground State Structure



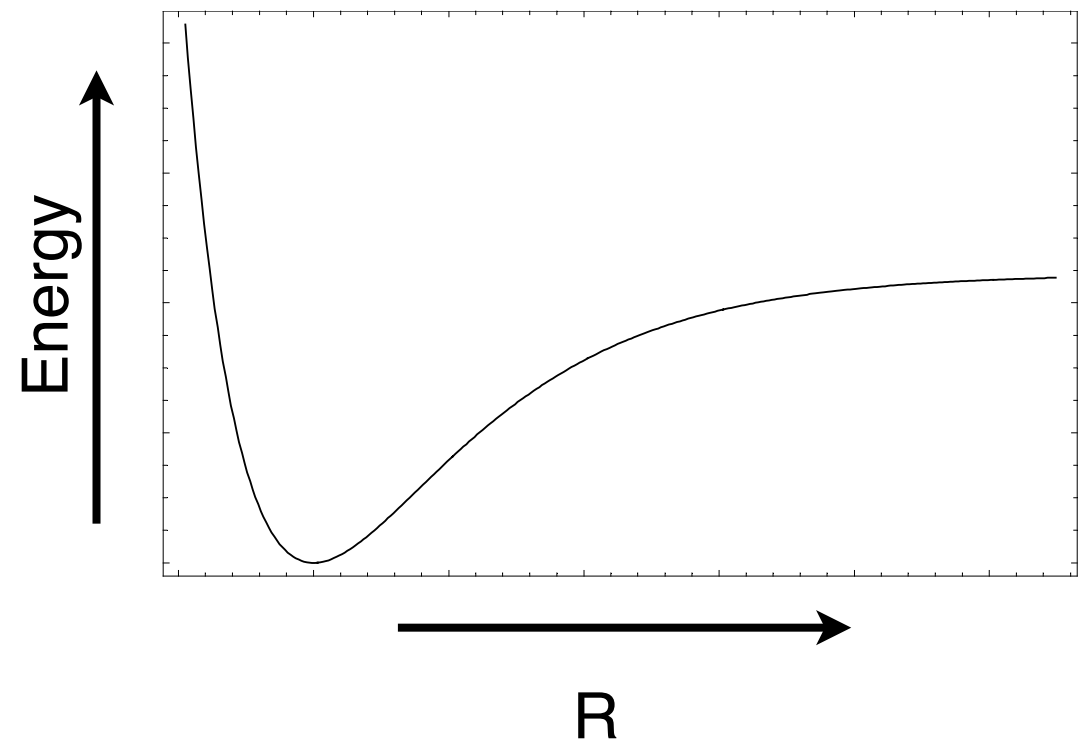
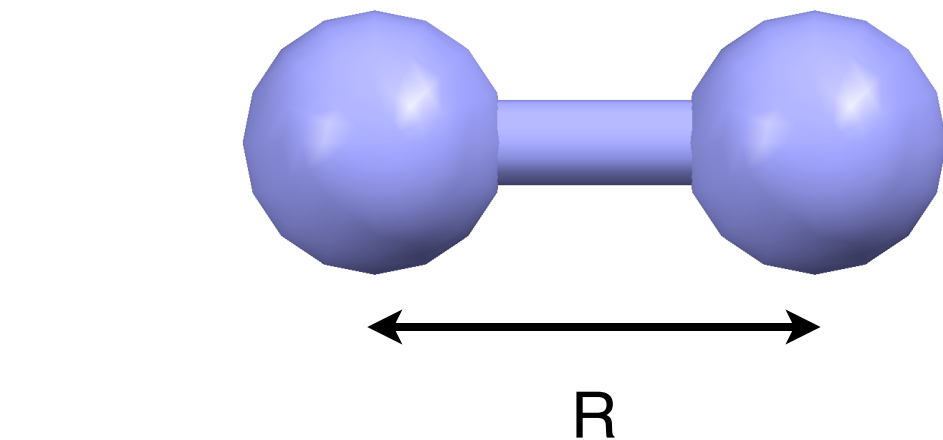
Forces

For a given set of atomic positions the ions will experience a force

$$F = \frac{dE}{dR}$$

After minimising the energy of the electrons
CASTEP can compute the ionic forces
(v. cheap using Hellman-Feynman theorem)

```
***** Symmetrised Forces *****
*
*      Cartesian components (eV/A)
*  -----
*      x      y      z
*
* N      1      0.56342      0.00000      0.00000
* N      2     -0.56342      0.00000      0.00000
*
*****
```



Rule of Thumb Force > 1.0 eV/Å is a large force

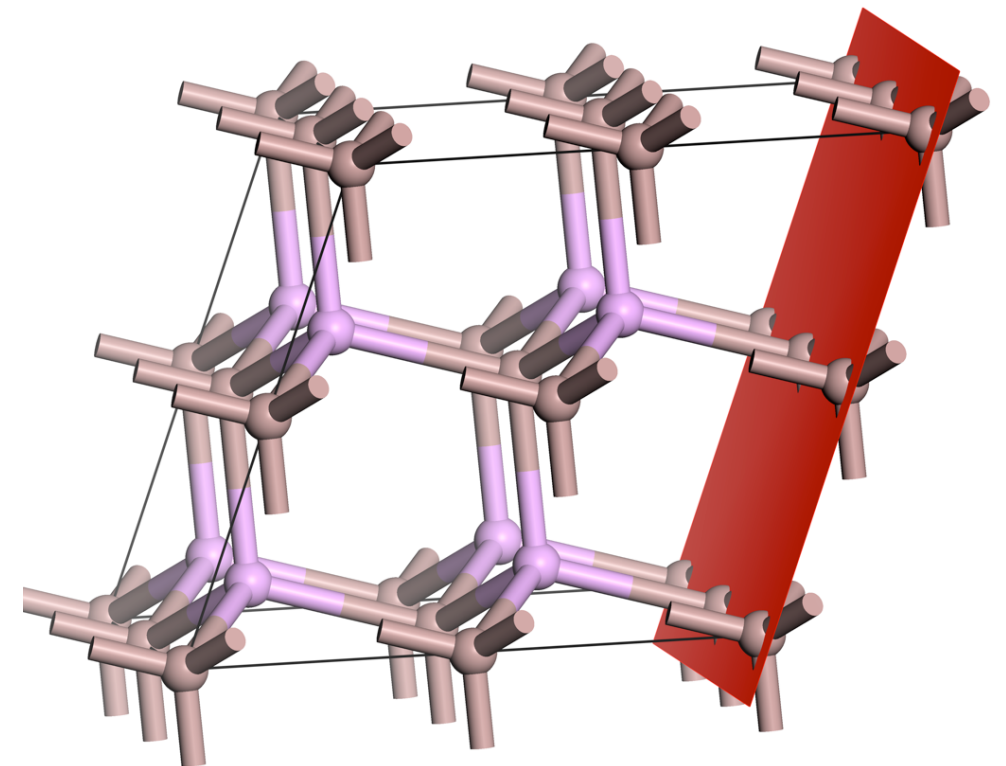
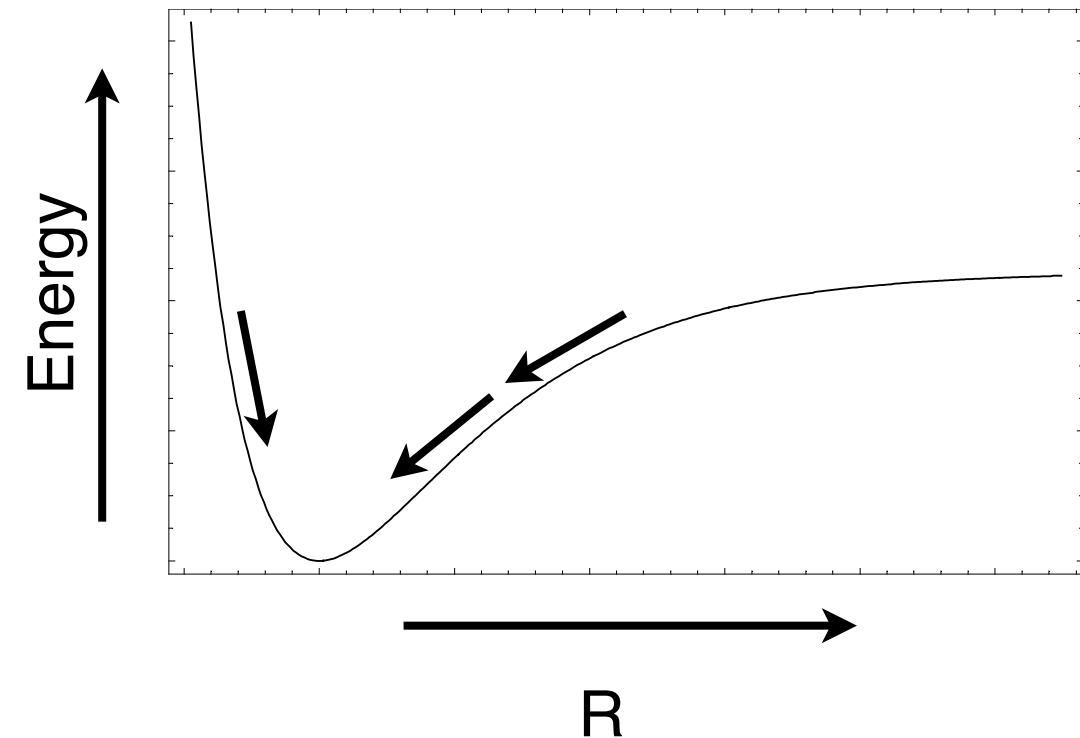
$$1 \text{ Ry/a.u.} = 25.7110 \text{ eV/Å}$$

Crystals

Ground-state structure at zero Kelvin corresponding to a minimum in the energy (all ionic forces are zero)

If we consider a crystalline material ground-state structure is a minimum in the Enthalpy (zero force and stress)

We can use the forces, and stress to guide us to the minimum



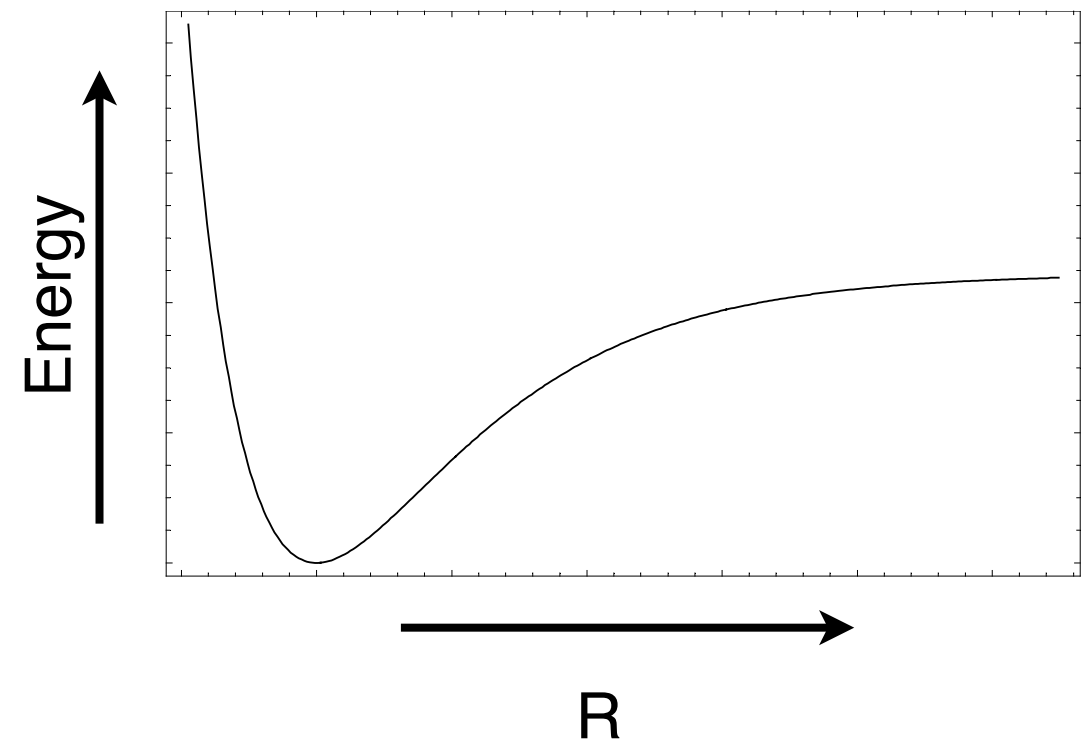
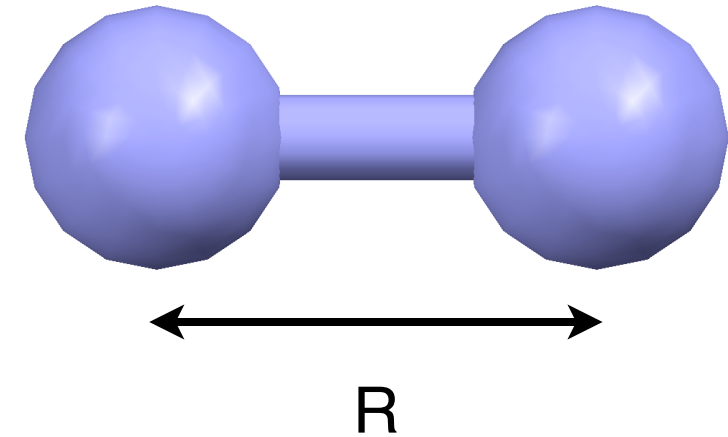
Geometry Optimisation

To find ground-state of system

- 1- Minimise energy of electrons for fixed ionic positions
- 2- Evaluate forces on ions (+stress)
- 3- Move ions, unit cell according to LBFGS algorithm

Repeat steps 1-3 until the maximum force / stress is below a specified value (plus a few other conditions)

Important note: LBFGS will find the closest local minimum - this might not be the global minimum



Constraints

We can apply constraints to system during minimisation

- Fix positions of an atom or a set of atoms (for example fix heavy atoms and only allow H atoms to move)
- Fix unit cell (all or only allow given lattice vectors or angles to change)
- Fix centre of mass
- Apply a pressure to the unit cell: for example to find structure at given pressure - study phase changes

Note: We can also fix bond lengths, or bond angles. These are 'non-linear constraints and so harder to apply.

Design your computational experiment

Need to think carefully about your problem

BFGS will find nearest minima.

Is this what you want?

Maybe there are multiple minima

Can compare total energy of minima to find the lowest

If the energies are close these might all be physically relevant

Does your system have some disorder?

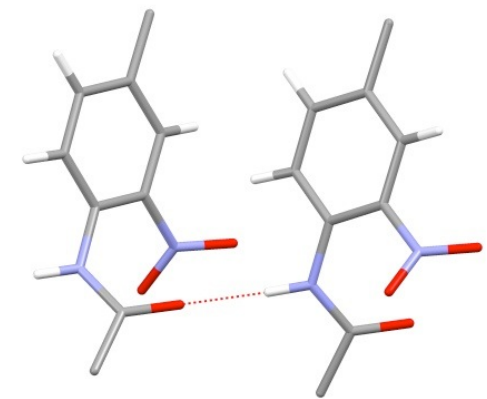
Is it static disorder

Is there a time-scale involved

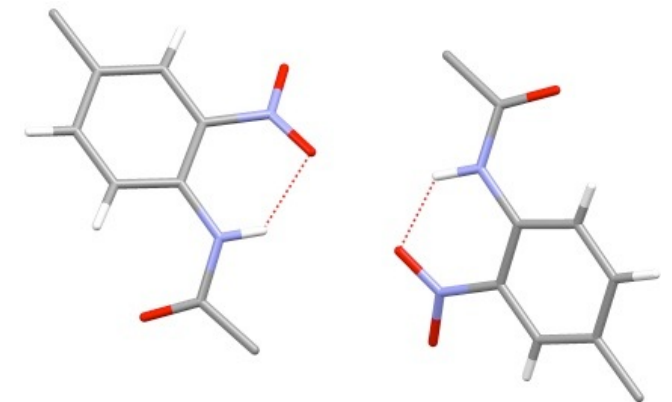
Minimisation will give the 0K structure.

A diffraction experiment would have been performed at a finite temperature.

A spectroscopic measurement might have been made at a different temperature.



Two polymorphic forms of MNA
both are stable at rtp



The ground-state energies of the
systems are close. But the
spectroscopic signatures (NMR,
vibrational) are quite different.

Starting Structures

Diffraction: Structures deposited in the ICSD (inorganic) and CSD (organic)

X-ray Diffraction: Many structures available (and equipment is common), but hydrogen positions inaccurate

Neutron Diffraction: Good resolution for light elements - but few structures available

For symmetric molecules or 1 or 2 dimensional systems (eg nanotubes): work out by hand, or use a programs such as tubegen.

Comparison of XRD & NRD

Sucrose - 3 structures in CSD

- **NRD** Neutron diffraction, G.M. Brown and H.A. Levy *Acta Cryst. B***29**, 790-797 (1973).
- **XRD-91** X-ray diffraction, R. C. Hynes Y. Le Page , *J. App. Cryst.*, 24 (1991) p352
- **XRD-73** X-ray diffraction, J. C. Hanson L. C. Sieker L. H. Jensen , *Acta Crysta, B*, 29 (1973) p797

Compare Forces (eV/Å) for Expt and Partial Relaxed (PR = only H atoms can move) structures

		NRD		XRD-91		XRD-73	
		Expt	PR	Expt	PR	Expt	PR
Hydrogen	Mean Force	0.6065		7.1615		12.7606	
	Maximum Force	3.8200		20.7193		66.0071	
Heavy Atoms	Mean Force	0.6028		6.4252		11.4669	
	Maximum Force	3.9187		19.9329		65.7079	

NRD max force (3.8eV/Å) on O-H of length = 0.912 Å

XRD-73 max force (66eV/Å) on O-H of length = 0.636 Å

Comparison of XRD & NRD

Sucrose - 3 structures in CSD

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Compare Forces (eV/Å) for Expt and Partial Relaxed (PR = only H atoms can move) structures

		NRD		XRD-91		XRD-73	
		Expt	PR	Expt	PR	Expt	PR
Hydrogen	Mean Force	0.6065	0.0011	7.1615	0.0014	12.7606	0.0037
	Maximum Force	3.8200	0.0027	20.7193	0.0036	66.0071	0.0081
Heavy Atoms	Mean Force	0.6028	0.2613	6.4252	0.3285	11.4669	0.2028
	Maximum Force	3.9187	0.4565	19.9329	0.9787	65.7079	0.3764

NRD max force (3.8eV/Å) on O-H of length = 0.912 Å

Relaxed bond length = 0.977 Å

XRD-73 max force (66eV/Å) on O-H of length = 0.636 Å

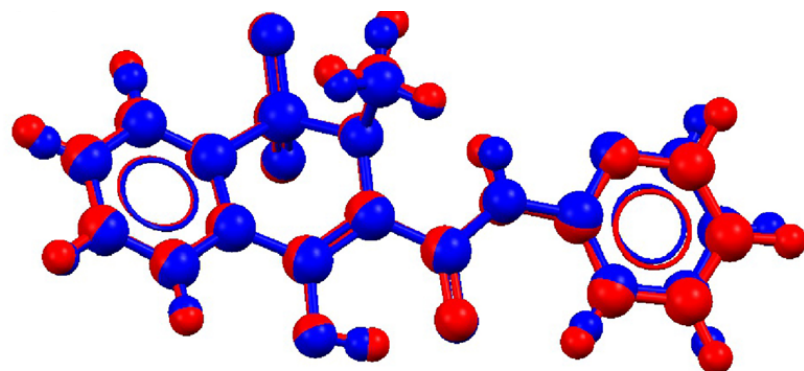
Note: We kept the unit cell fixed during the relaxation. This is important when using standard functions LDA/GGA as they do not describe dispersion/Van der Waals forces well - and these have a large contribution to the cohesion of molecular crystals.

Piroxicam

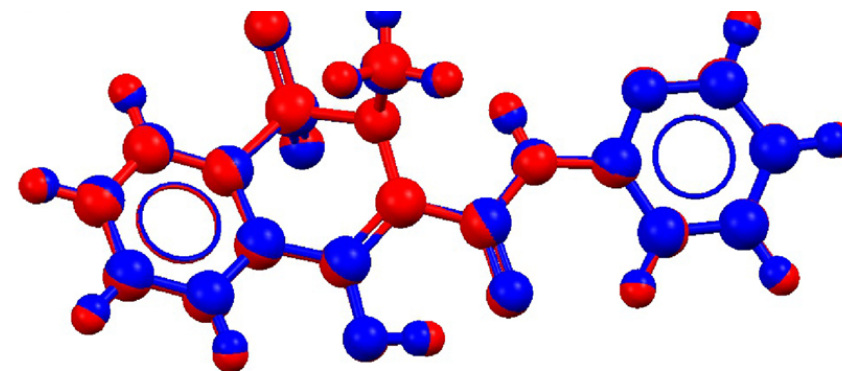
Improving Confidence in Crystal Structure Solutions Using NMR Crystallography

Before geom opt
After geom opt

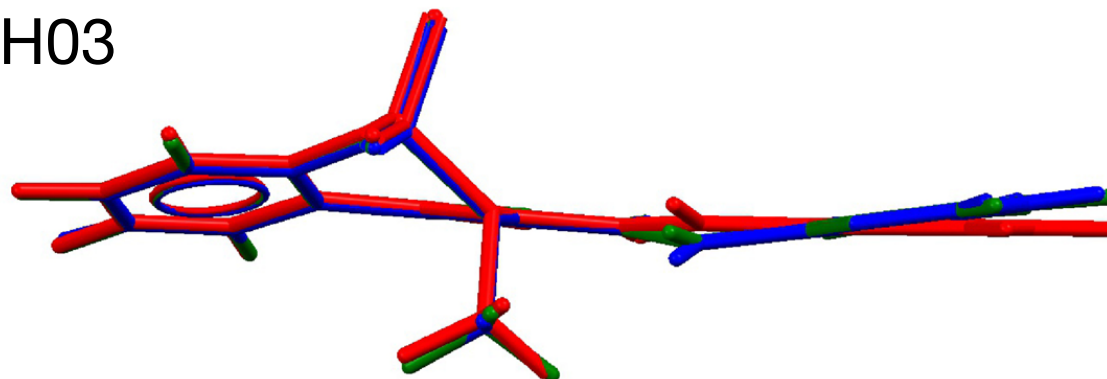
BIYSEH03 PXRD R=19%



BIYSEH13 SXRD R=2.4%



BIYSEH03



Andrew Tatton + AstraZenica + CCP-NC
Cryst. Growth Des. 2018, 18, 3339

Piroxycam

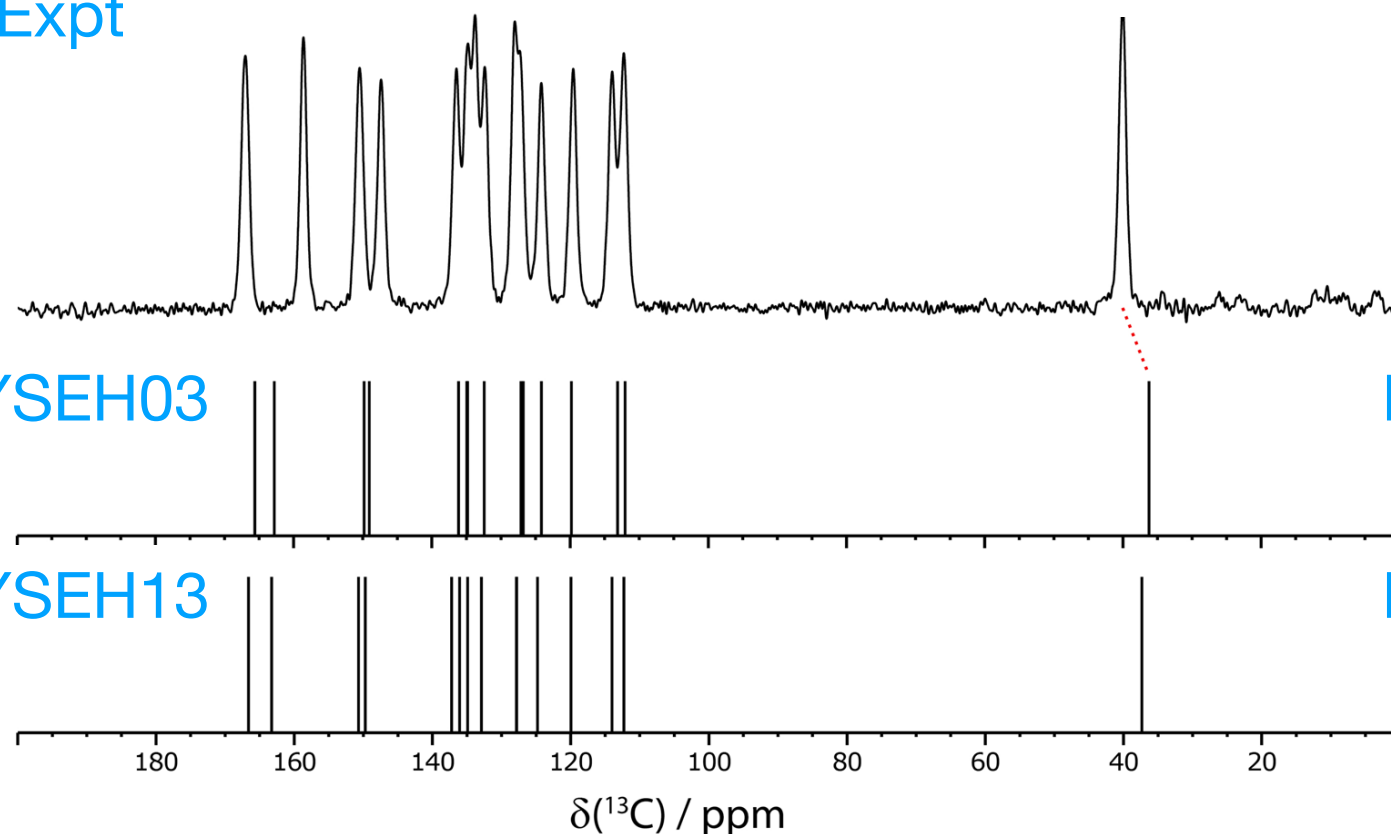
Improving Confidence in Crystal Structure Solutions Using NMR Crystallography

^{13}C

Expt

BIYSEH03

BIYSEH13

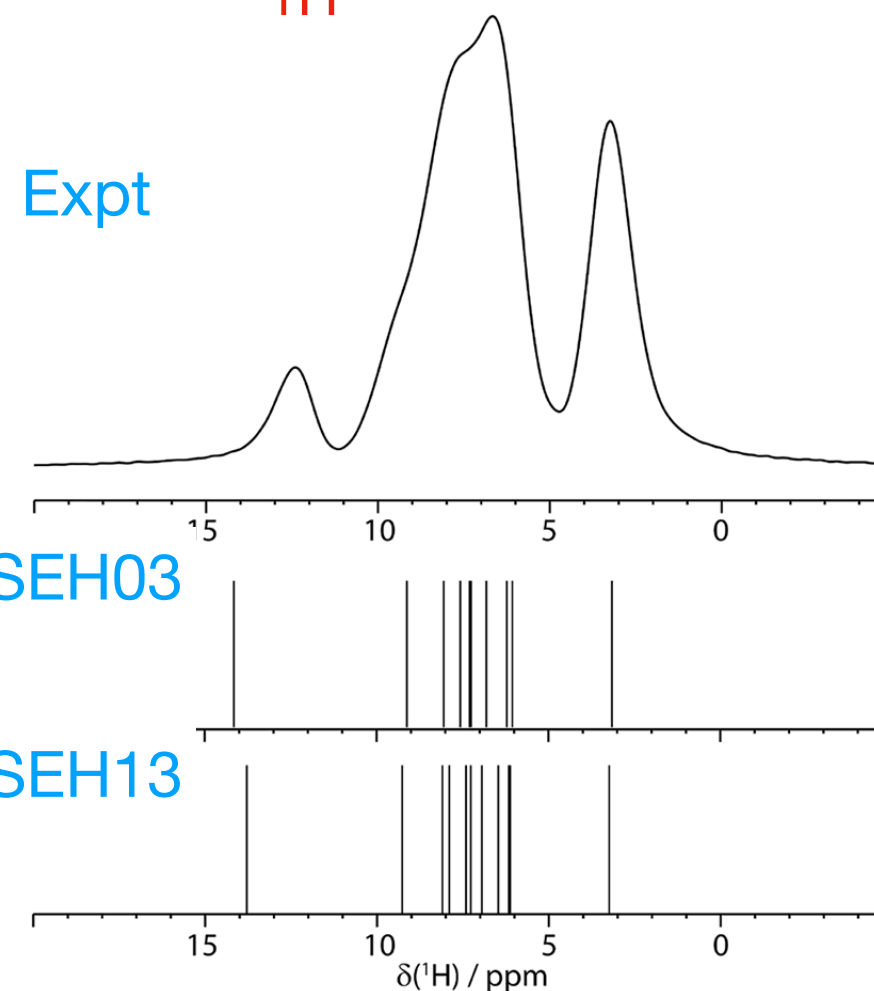


^1H

Expt

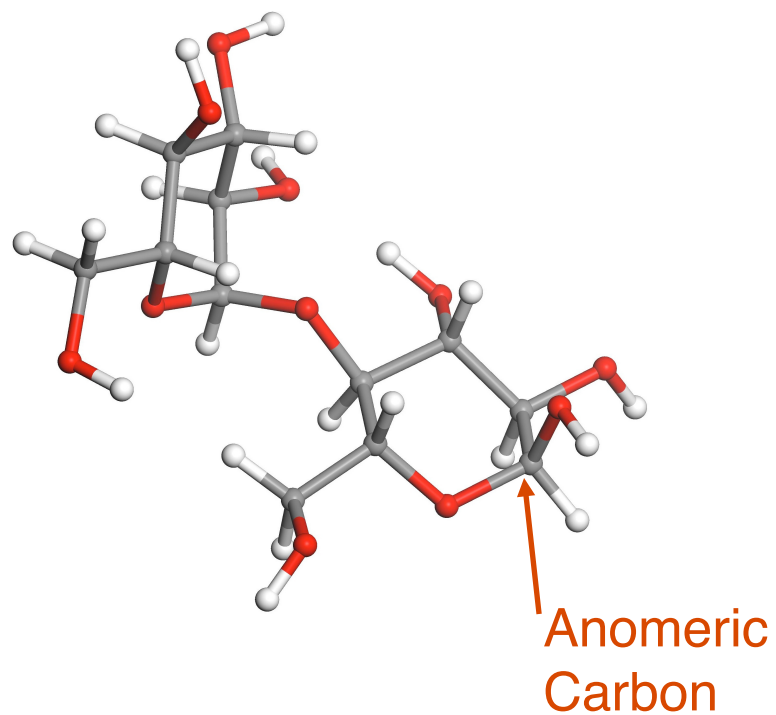
BIYSEH03

BIYSEH13



Both structural models (after geometry optimisation) provide equivalent description of NMR parameters

Maltose

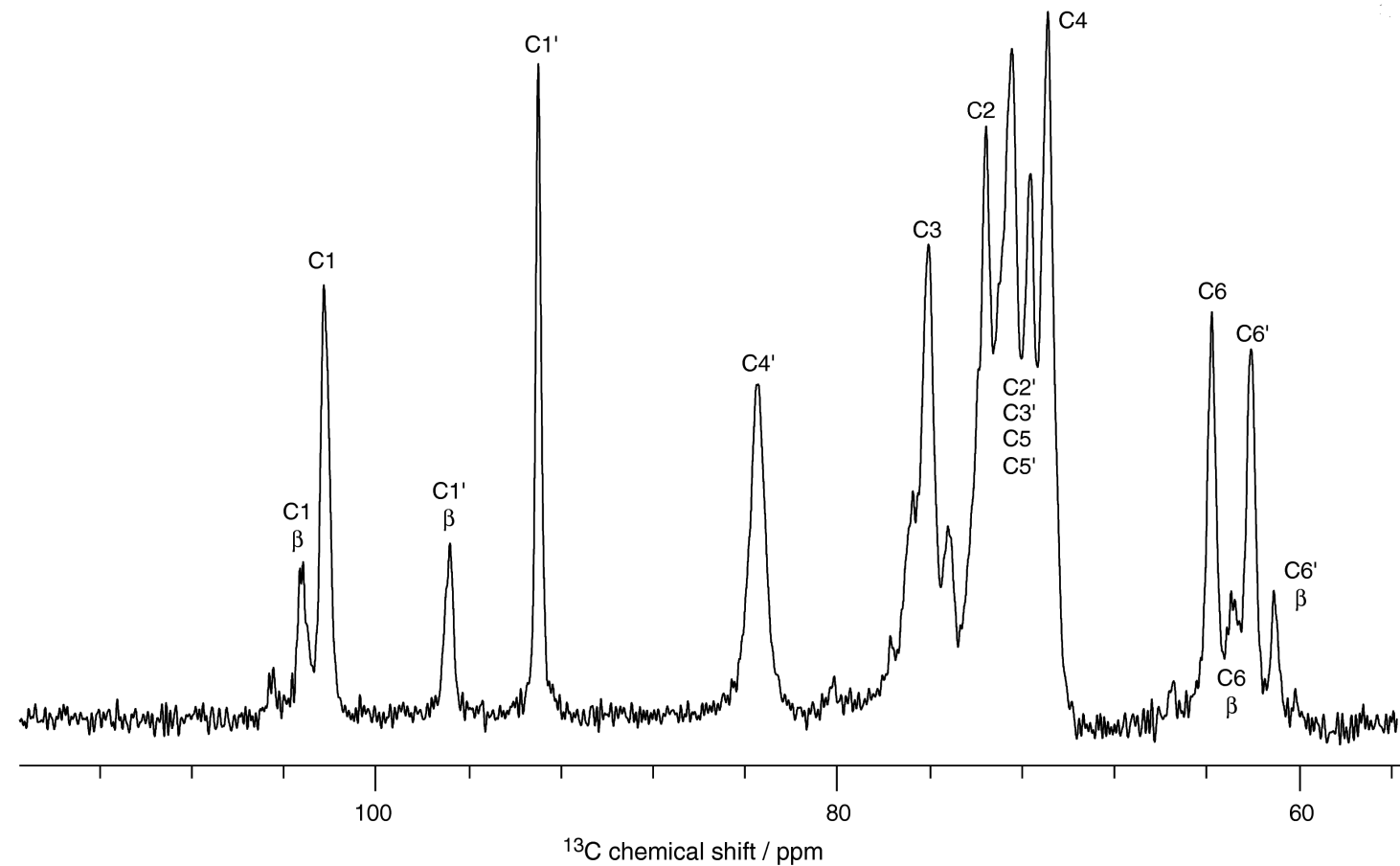


Crystallises as:

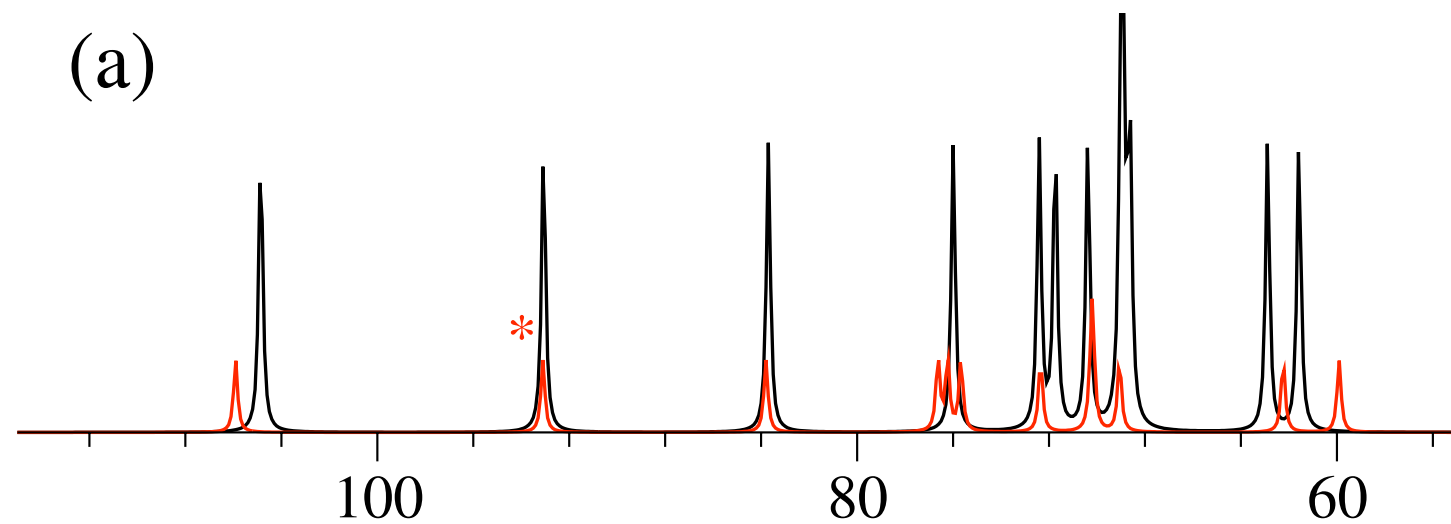
80% α -Maltose

20% β -Maltose

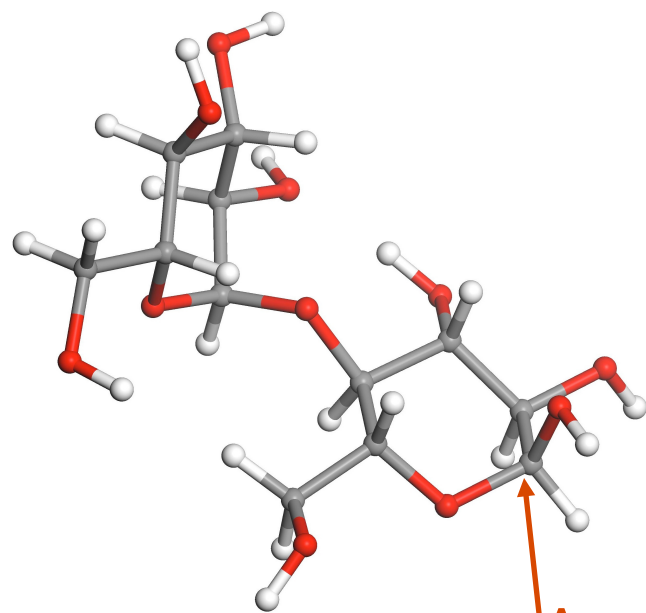
Compute spectra for both structures separately and superimpose



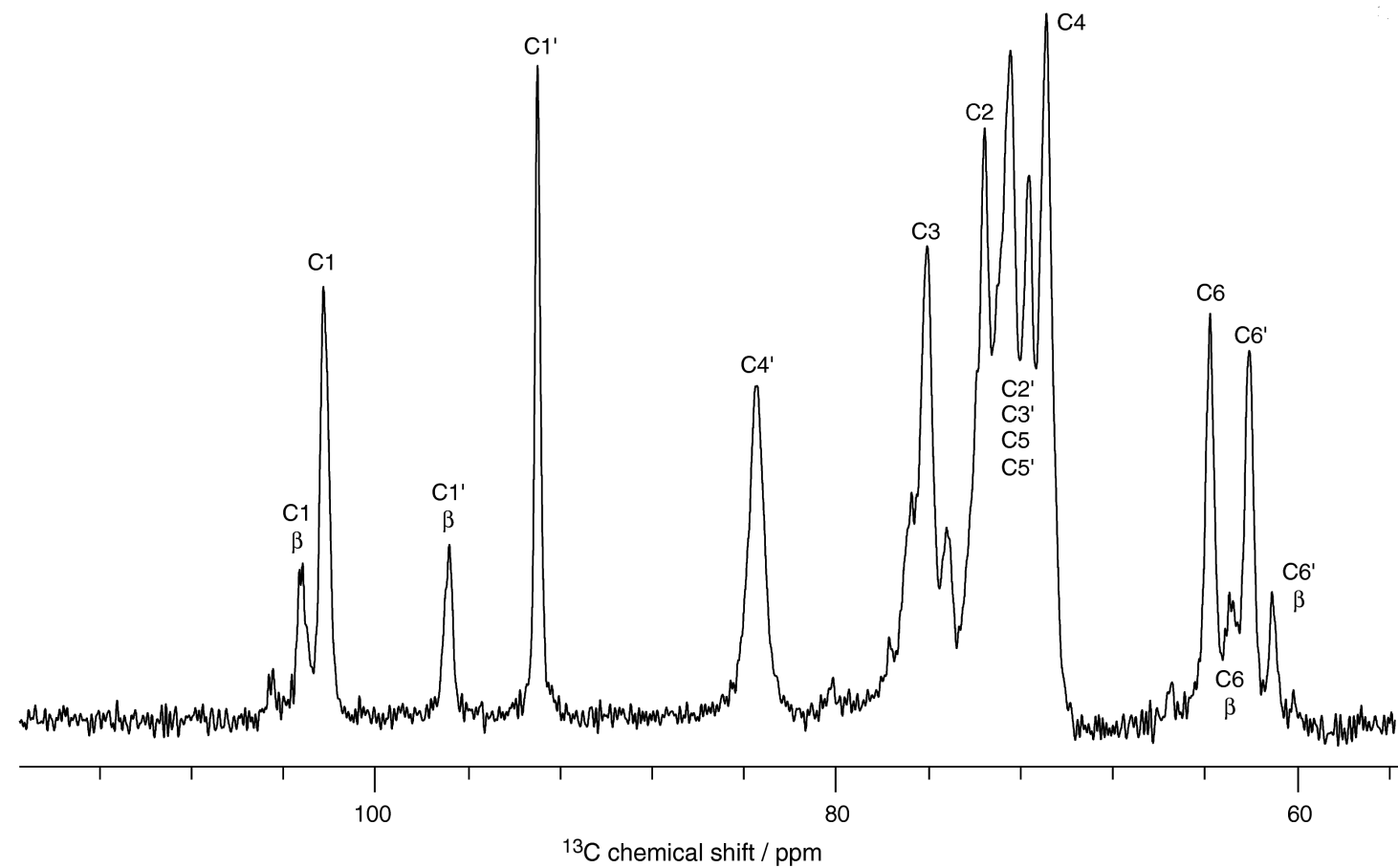
(a)



Maltose

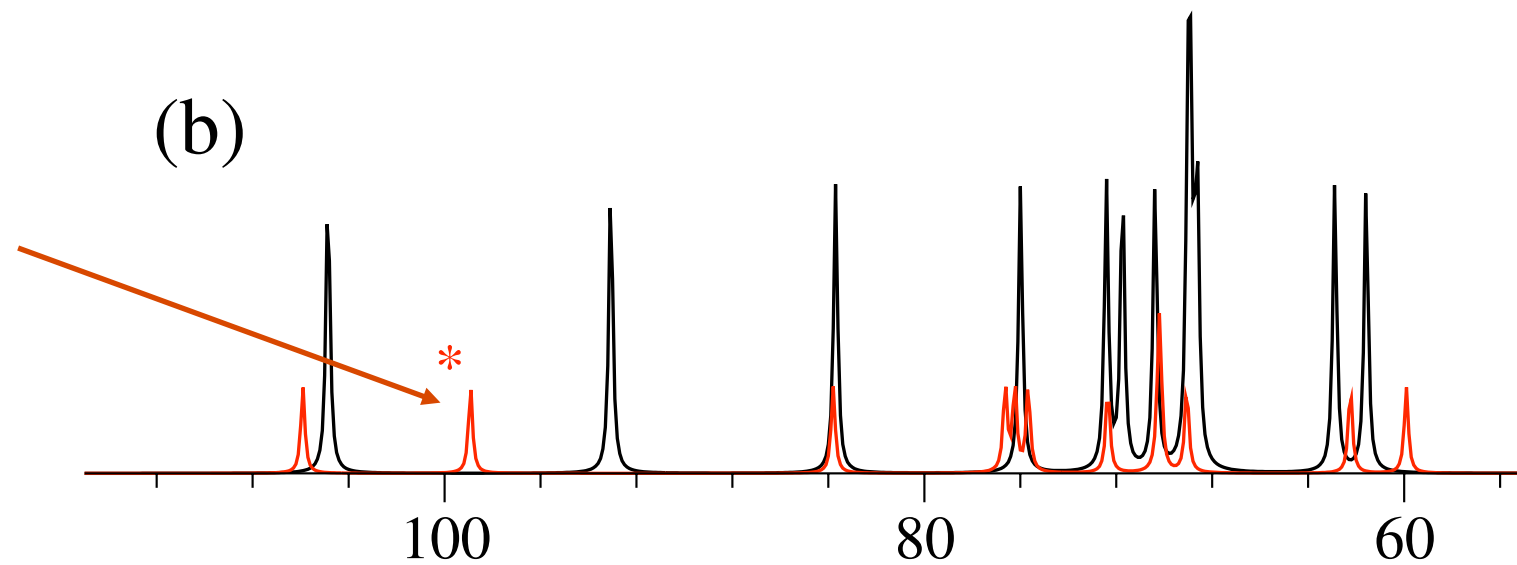


Anomeric
Carbon



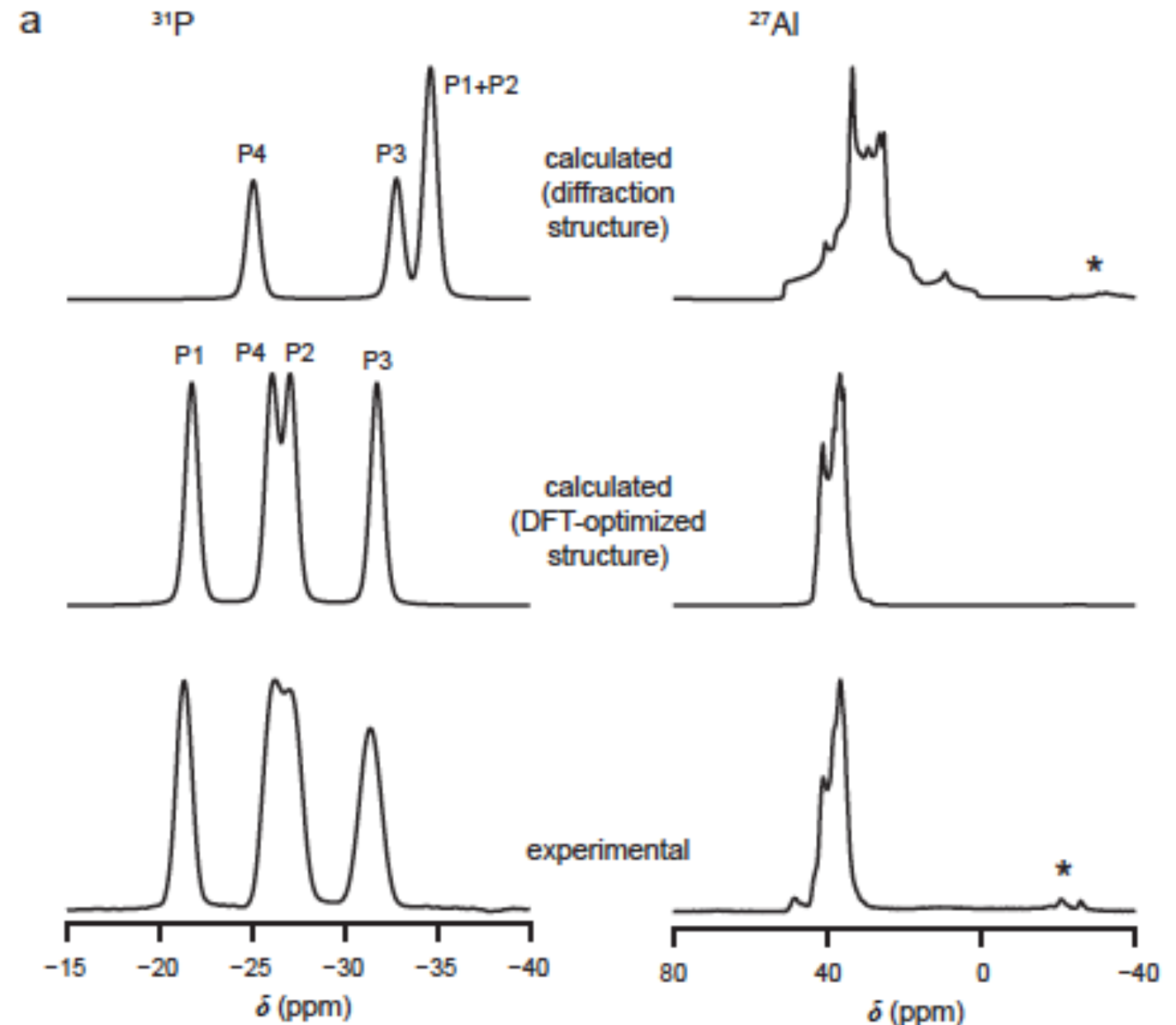
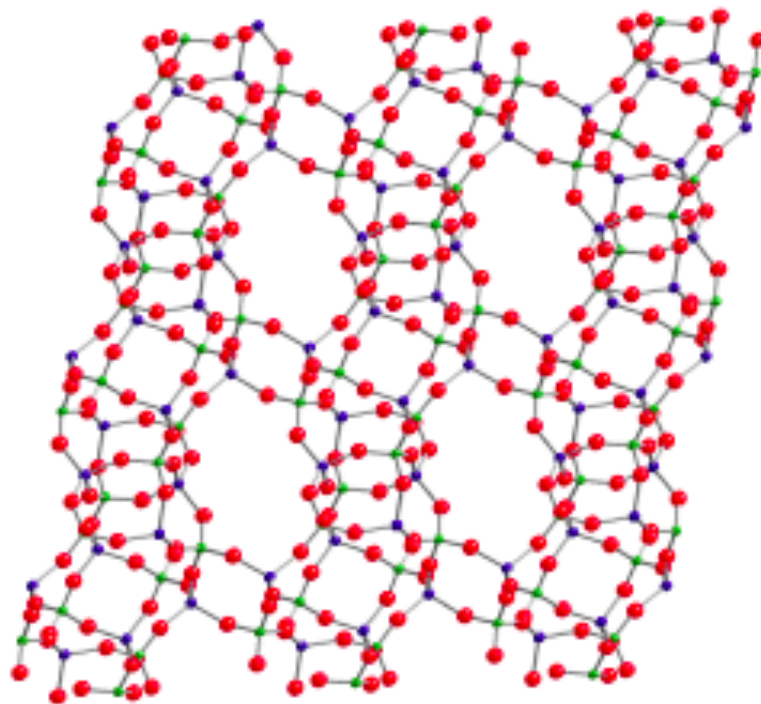
After relaxing neighbouring atom O1'
which had large force / thermal ellipsoid

(b)



Aluminophosphate

- Microporous aluminophosphate (AIPO) frameworks have applications in gas storage and separation, catalysis and drug delivery
- AIPO-14 has the AFN framework with 4 P and 4 Al sites expected



slide from Sharon Ashbrook (St Andrews)

*.param keywords

```
task : GeometryOptimization

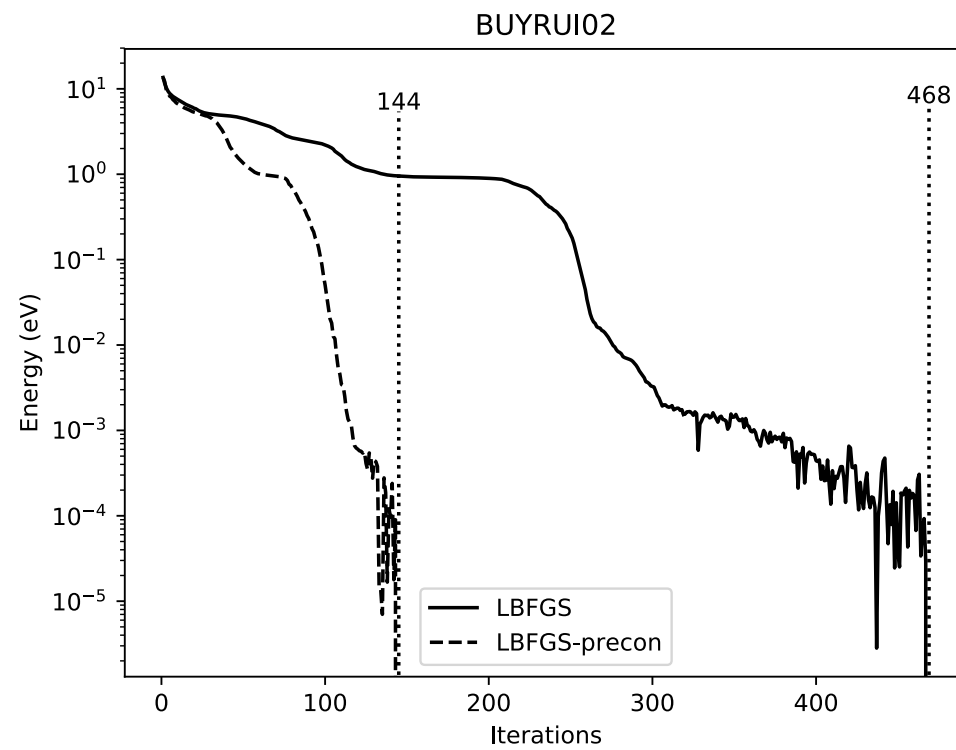
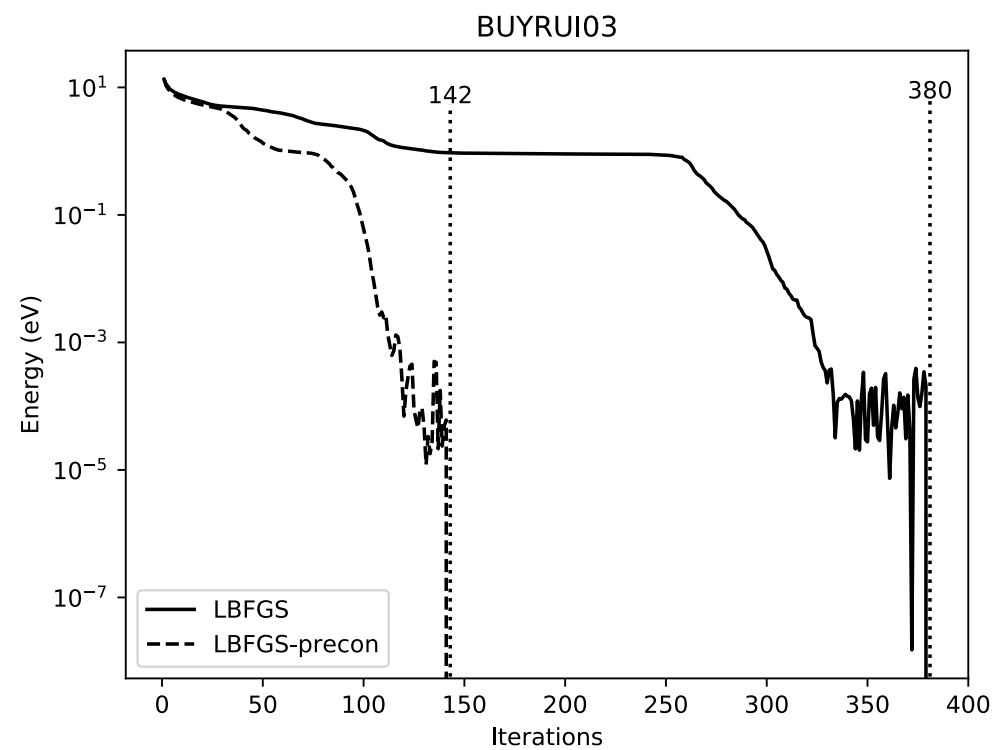
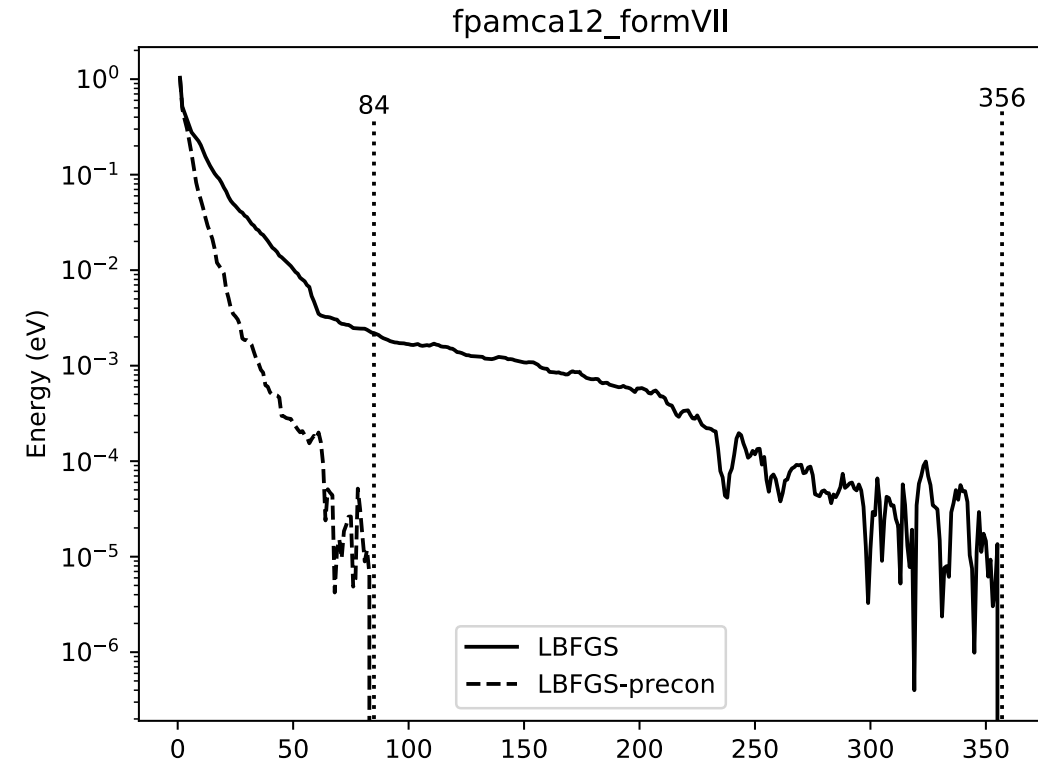
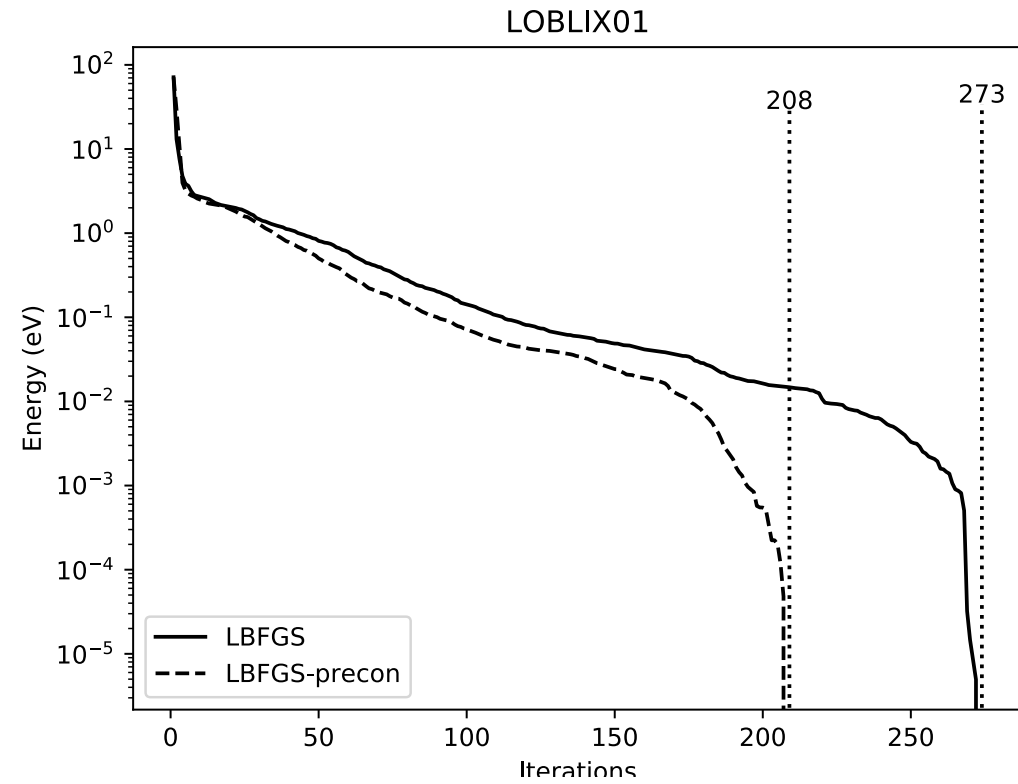
geom_method      : LBFGS (default), BFGS, DAMPEDMD,TPSD

geom_preconditioner = EXP / FF / ID
                  # EXPonential, Force Field, Identity (default)

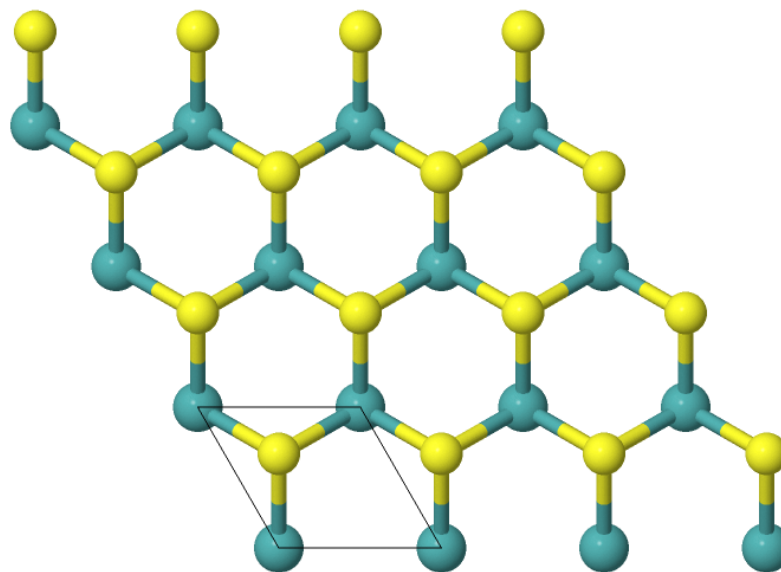
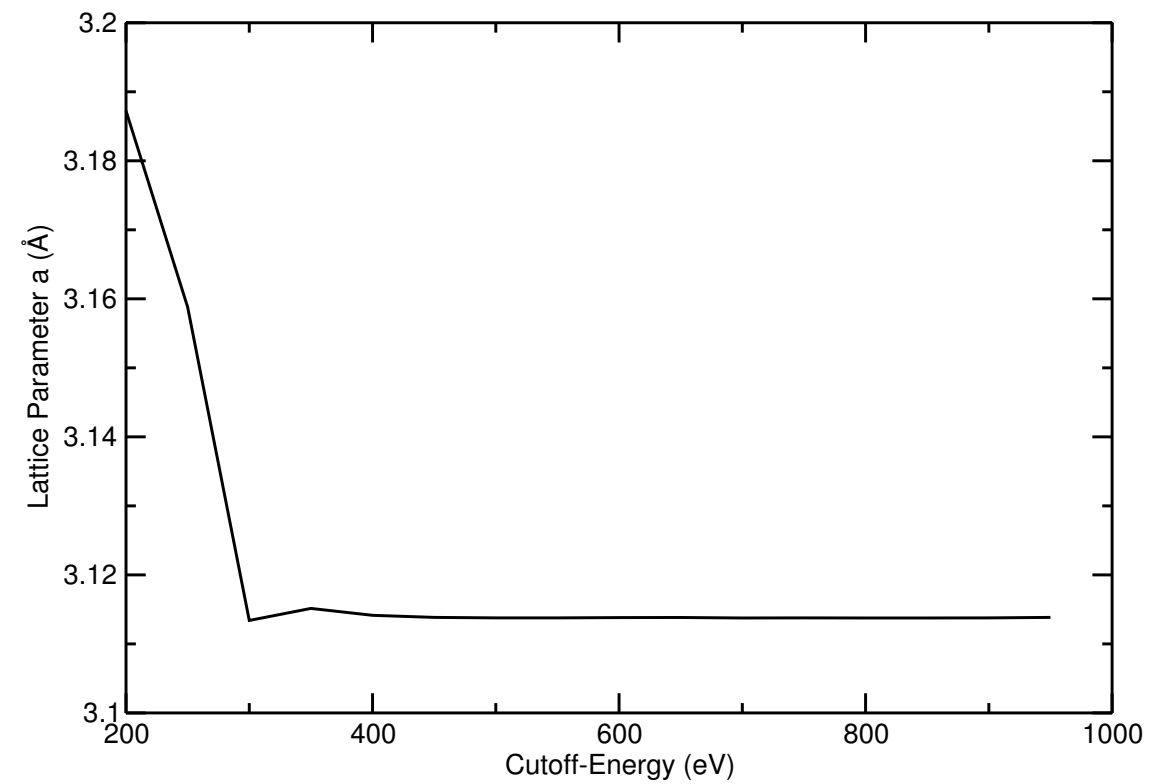
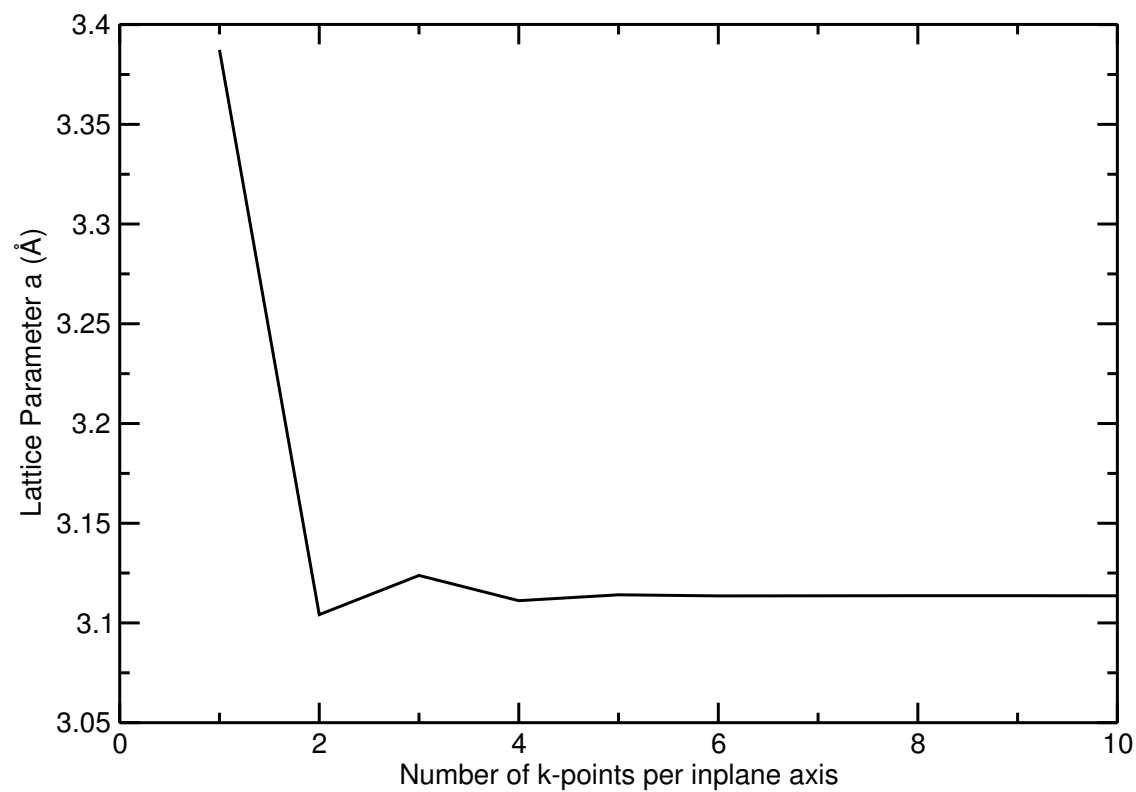
geom_energy_tol  : 2.0e-5  5.00e-6  eV
geom_force_tol   : 0.05    0.01     eV/ANG
geom_stress_tol  : 0.1     0.02     GPa
geom_disp_tol    : 0.001   5.00e-004 Ang
geom_max_iter    : 30      200

write_cell_structure : T
write_cif_structure  : T
```

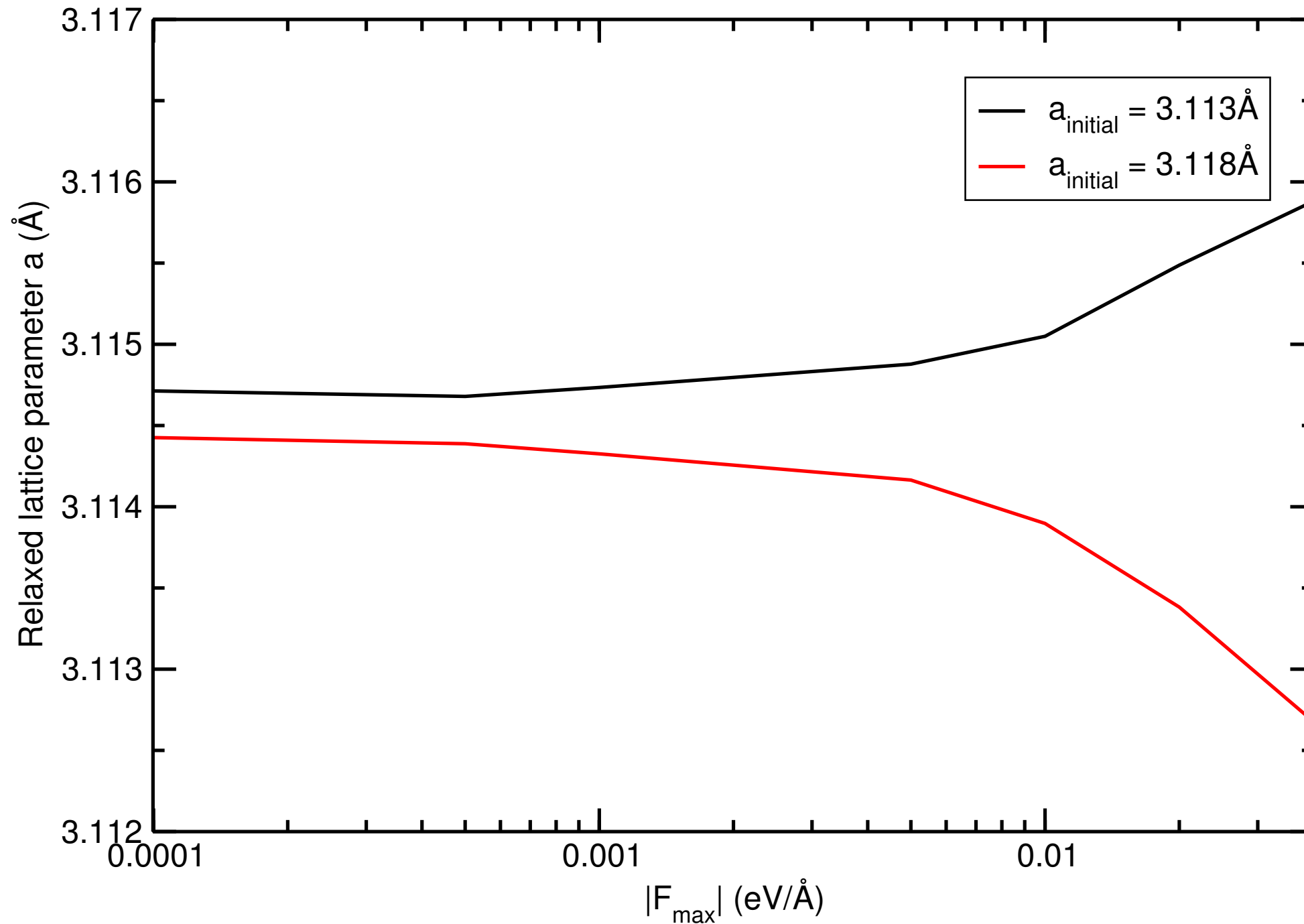
Preconditioning



Convergence - kpoints and basis-set



Convergence - geom_force_tol



*.cell keywords

```
FIX_CELL : T / F  
FIX_VOL  : T / F
```

```
%BLOCK CELL_CONSTRAINTS  
      1      2      3  
      0      0      0  
%ENDBLOCK CELL_CONSTRAINTS
```

a, b, c can vary, cell angles fixed

```
%BLOCK CELL_CONSTRAINTS  
      1      1      2  
      0      0      0  
%ENDBLOCK CELL_CONSTRAINTS
```

a = b, c can vary, cell angles fixed

Note: Variable cell calculations need good basis set convergence. By default CASTEP will use a finite basis-set correction to improve the calculation of the stress. If the starting and final unit cells differ significantly, consider starting a fresh geometry optimisation from the final structure.

*.cell keywords

all versions of castep

FIX_ALL_IONS : T / F

```
%BLOCK IONIC_CONSTRAINTS
  1      W      1      1.0000000000      0.0000000000      0.0000000000
  2      W      1      0.0000000000      1.0000000000      0.0000000000
  3      W      1      0.0000000000      0.0000000000      1.0000000000
  4      W      2      1.0000000000      0.0000000000      0.0000000000
  5      W      2      0.0000000000      1.0000000000      0.0000000000
  6      W      2      0.0000000000      0.0000000000      1.0000000000
  7      W      3      1.0000000000      0.0000000000      0.0000000000
  8      W      3      0.0000000000      1.0000000000      0.0000000000
  9      W      3      0.0000000000      0.0000000000      1.0000000000
 10      W      4      1.0000000000      0.0000000000      0.0000000000
 11      W      4      0.0000000000      1.0000000000      0.0000000000
 12      W      4      0.0000000000      0.0000000000      1.0000000000
%ENDBLOCK IONIC_CONSTRAINTS
```

shortcuts in castepv17+

fix: all	fix all atoms
fix: C N	fix species C and N
fix: C 1	fix position of atom C 1
fix: C{1,3,5-10}	fix positions of atoms C1, C3, C5,C6,C7,C8,C9,C10
fix: all unfix: H	fix positions of all atoms except H